# Large Language Models in Medicine and Healthcare: Capabilities, Evaluation, Workflows, and Governance

Paperclip–Codex survey workflow

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**Abstract**

Large language models (LLMs) have moved from biomedical text-mining tools to general-purpose clinical assistants, open medical models, retrieval-augmented systems, multimodal agents, and workflow tools for documentation, patient messaging, trial matching, and decision support. This survey synthesizes 103 structured evidence cards drawn from 1,020 screened candidates found through four Paperclip search rounds and integrated web search. The literature shows three simultaneous trends. First, domain adaptation still matters: biomedical encoders, clinical-note models, structured-EHR transformers, and EHR-scale models each expose distinct medical data distributions [39] [18] [92] [85] [40] [62]. Second, benchmark performance has improved rapidly, from early ChatGPT-on-USMLE studies and Med-PaLM to HealthBench, MedHELM, MedXpertQA, MedBench, GMAI-MMBench, and LiveMedBench; however, exam scores, static QA, VQA scores, and rubric scores remain proxies rather than clinical outcome evidence [69] [1] [49] [100] [35] [76] [44] [7] [91]. Third, the most clinically plausible near-term uses are assistive and reviewed workflows: retrieval-grounded answers and trial matching, simulated diagnostic dialogue, patient-message drafts, ambient documentation, radiology report drafting, and administrative extraction all have growing but task-specific evidence [95] [82] [30] [72] [9] [47] [79]. We conclude that medical LLMs should be evaluated as sociotechnical clinical systems rather than as isolated text generators: deployment requires task-specific validation, human oversight, uncertainty and equity assessment, adversarial testing, privacy protection, lifecycle monitoring, and governance matched to intended use [61] [57] [13] [55] [66] [73] [54].

The figure summarizes the survey's organizing logic: biomedical and clinical foundations feed general and medical LLMs, grounding and agents connect models to reviewed workflows, and clinical deployment depends on evaluation evidence, safety and equity controls, governance, and lifecycle monitoring.

## 1 Introduction

Medicine is unusually language-intensive. Diagnosis begins in conversation, care plans are encoded in notes and guidelines, clinical trials are matched through eligibility text, and patients increasingly seek medical explanations through written channels. LLMs therefore appear naturally suited to healthcare, yet this fit is also the source of danger: plausible language can mask factual error, missing clinical context, uncalibrated uncertainty, bias, or unsafe action. The current literature should be read as evidence of accelerating capability, not as a single verdict that LLMs are ready for autonomous care [56] [4] [46].

The field did not begin with ChatGPT. Biomedical and clinical NLP had already shown that domain corpora matter. BioBERT and PubMedBERT adapted transformer encoders to biomedical literature, while ClinicalBERT and GatorTron targeted clinical notes and EHR narratives [39] [23] [18] [92]. BioGPT then demonstrated the value of biomedical generative pretraining for text generation and mining [48]. These models mostly addressed extraction, prediction, and benchmark QA rather than open-ended clinical dialogue, but they established the central

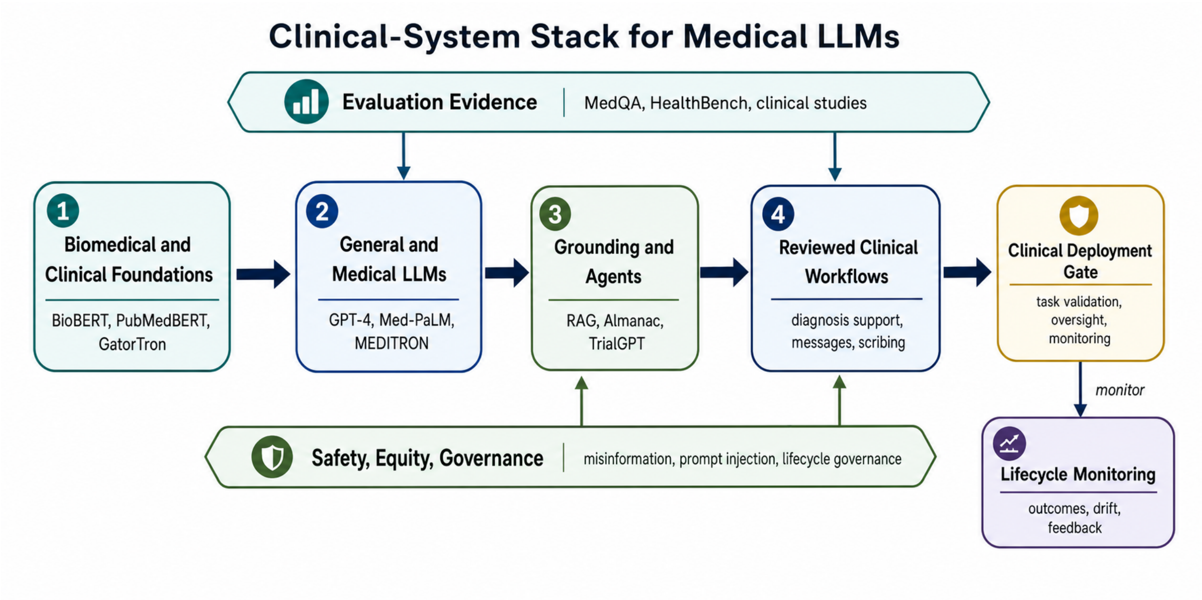


Figure 1: Overview of LLMs in medicine and healthcare as a layered clinical-system stack.

premise of medical LLM work: general language competence is insufficient when terminology, evidence standards, data privacy, and clinical workflow are domain-specific.

The post-2022 wave added instruction-following, chat interfaces, large-scale proprietary models, open medical LLMs, and multimodal systems. Med-PaLM and MultiMedQA made medical QA evaluation multidimensional by combining multiple datasets with human assessment of factuality, consensus, harm, bias, and helpfulness [69]. GPT-4 then showed that a general-purpose frontier model could perform strongly on USMLE-style and medical challenge problems without medical fine-tuning [53]. Med-PaLM 2, HealthBench, MedHELM, and MedXpertQA pushed the evaluation frontier toward expert-level QA, open-ended rubric evaluation, broader task suites, and multimodal expert reasoning [70] [1] [49] [100]. Yet these studies also clarify the central tension of the field: medical benchmarks are useful for measuring progress, but they are not equivalent to safe clinical work.

This survey therefore asks not only which models perform well, but which kinds of evidence support which kinds of claims. A model can be strong on MedQA and still fail at investigation or treatment choices, sequential information gathering, or realistic EHR actions [33] [26]. A patient-message or ambient-scribe system can reduce burden while creating new questions about tone, review responsibility, note quality, well-being, and equity [3] [72] [47] [79]. A retrieval-augmented or multimodal system can improve grounding but still fail when retrieved context is insufficient, when medical misinformation corrupts behavior, or when prompt injection reaches a vision-language setting [95] [90] [88] [17]. For these reasons, the survey is organized around functional roles, evidence types, and deployment risks rather than a leaderboard of model names.

## 2 Methods/Search Strategy

This review used a narrative-survey workflow with scoping-review discipline. Paperclip retrieval was run in four rounds. Round 1 used 26 broad and targeted queries covering medical LLMs, ChatGPT/GPT-4, medical QA, biomedical and clinical language models, documentation, RAG, multimodality, privacy, fairness, safety, and governance. Round 2 added 18 gap-driven queries for AMIE, HealthBench, ambient scribes, patient portal messages, TrialGPT, Almanac, data poisoning, health equity, and named multimodal/open medical systems. A web-search audit then identified missing landmarks, leading to a focused third Paperclip round for MedHELM, MedXpertQA, NYUTron, clinical decision-making limitations, patient-inbox deployments, randomized diagnostic-reasoning trials, medical agents, and AI co-scientist systems. A coverage-auditor critique triggered a fourth focused round for open-weight clinical models, Chinese and multilingual medical LLMs, structured-EHR foundation models, medical VQA/radiology benchmarks, EHR agents, ambient-scribe trials, hallucination/calibration benchmarks, privacy/de-identification, HIPAA de-identification guidance, and benchmark contamination.

Codex built-in web search was used as an integrated recall and verification layer, not as a substitute for evidence extraction. It verified publication status, added official governance sources, located later peer-reviewed versions of preprints, and identified landmarks outside Paperclip's strongest full-text coverage. Examples include the Nature and Nature Medicine versions of Med-PaLM and Med-PaLM 2, the Nature articles on AMIE, the Nature Communications version of TrialGPT, the WHO large-multimodal-model guidance, FDA AI/ML Software as a Medical Device materials, and NIST AI Risk Management Framework resources [69] [70] [82] [50] [30] [57] [13] [55].

After deduplication by normalized title and merging across discovery sources, the integrated candidate table contained 1,020 records. Screening used four tiers: `core`, `adjacent`, `context`, and `out_of_scope`. Preprint status was recorded but was not used as an exclusion reason. The final evidence base contains 103 structured evidence cards. Because the candidate pool exceeded 150 records, the evidence-card count exceeds the project threshold of at least 50 cards and was expanded after coverage audit to improve representativeness across open models, multimodal systems, EHR modeling, deployment evidence, and safety. The manuscript cites only papers or guidance documents represented by evidence cards.

# 3 Taxonomy

## 3.1 1. Biomedical and Clinical Language-Model Foundations

The first layer is domain adaptation. Biomedical encoders such as BioBERT and PubMedBERT showed that pretraining on biomedical text improves biomedical NLP [39] [18]. ClinicalBERT and GatorTron showed that clinical-note modeling is a distinct problem, with GatorTron scaling clinical language modeling over large deidentified clinical text corpora [23] [92]. BioGPT and GatorTronGPT extended this lineage toward GPT-style generation over biomedical literature and clinical text [48] [93]. A separate structured-EHR lineage includes BEHRT, Med-BERT, CEHR-BERT, EHRSHOT/CLMBR, and NYUTron, which model coded or note-based patient histories for prediction rather than open-ended dialogue [85] [40] [62] [60] [25]. The technical lesson is that "medical LLM" is not one thing. Literature models, note models, structured-EHR models, chat assistants, and predictive encoders differ in data, objectives, and risks.

## 3.2 2. Medical QA and Benchmark-Centered Systems

Medical QA benchmarks provide the most visible performance narrative. MedQA, PubMedQA, and MedMCQA became central because they test professional exam knowledge, biomedical abstract reasoning, and broad medical multiple-choice questions [29] [28] [58]. Early ChatGPT-on-USMLE studies made the educational stakes visible by showing that general conversational models could approach or exceed some licensing-exam thresholds while still varying by topic and difficulty [35] [14]. MultiMedQA combined benchmark datasets with consumer health questions and human evaluation, making Med-PaLM a landmark not only for performance but for evaluation design [69]. GPT-4 showed that a closed general-purpose model could perform strongly without medical training [53]. Recent benchmarks such as HealthBench, MedHELM, MedXpertQA, CMExam, MedBench, and LiveMedBench respond to saturation, language coverage, contamination, and ecological-validity concerns by adding open-ended conversations, broader task suites, expert reasoning, non-English medical exams, real-world clinical cases, and temporal separation [1] [49] [100] [44] [7] [91].

The limitation is not that these benchmarks are useless; it is that they answer only bounded questions. Multiple-choice accuracy measures recognition under artificially constrained options. PubMedQA tests abstract-grounded biomedical reasoning but not bedside care [28]. HealthBench improves realism through physician rubrics, but it remains a response-quality benchmark rather than a patient-outcome study [1]. LiveMedBench makes the contamination problem explicit by using continuously updated cases and temporal separation [91]. AgentClinic and MedAgentBench make a complementary point: when static medical knowledge is converted into sequential clinical interaction or EHR action, task performance can degrade and invalid actions become visible [67] [26].

### 3.3 3. Open and Domain-Specialized Medical LLMs

Open medical models respond to practical healthcare concerns about reproducibility, local validation, auditability, and settings where protected health information cannot be sent freely to closed systems. MedAlpaca, PMC-LLaMA, ChatDoctor, MEDITRON, MedGemma, BioMistral, Med42, Clinical Camel, HuaTuo, HuatuoGPT, DISC-MedLLM, and Apollo exemplify several strategies: continued pretraining on biomedical corpora, guideline/textbook incorporation, instruction tuning, dialogue-based knowledge encoding, doctor-data alignment, multilingual training, and parameter-efficient or full-parameter fine-tuning [8] [20] [86] [41] [77] [36] [11] [80] [83] [96] [75] [84]. MEDITRON is especially important because it scaled open medical pretraining to 70B parameters and emphasized guideline corpora as high-value medical data [8]. BioMistral and Med42-v2 show that the open ecosystem expanded beyond early LLaMA derivatives into Mistral- and Llama-3-based clinical model suites [36] [12].

Openness improves inspectability and local validation options, but it is not itself a safety guarantee. Open models may face data-quality limits, benchmark contamination, benchmark-centered evaluation, language and geography gaps, and insufficient deployment validation [8] [20] [86] [77] [91]. Some open medical LLMs are preprint-only or technical reports, and many are evaluated primarily on exam, QA, or consultation benchmarks rather than clinical outcomes [36] [11] [12] [96] [75]. Therefore, open medical LLMs should be framed as infrastructure for reproducible research and local validation, not as automatically deployable clinical products [63].

### 3.4 4. Grounded, Retrieval-Augmented, and Tool-Using Systems

A second technical response to medical risk is grounding. Almanac connected LLM answers to retrieved clinical sources and reported improved factuality and grounding relative to unsupported generation in an early clinical QA setting [95]. Broader RAG benchmarks show why this is necessary but insufficient: retrieval can improve medical QA, yet irrelevant or incomplete context can mislead generation, and systems must be evaluated for sufficiency, integration, and robustness [90]. Med-PaLM 2's retrieval and refinement methods reflect the same principle: hard clinical questions often require external evidence and claim checking rather than parametric recall alone [70].

Grounding is also moving into workflow systems. TrialGPT combines retrieval, criterion-level eligibility prediction, and trial ranking for patient-to-trial matching; its pilot user study reported reduced screening time, while explicitly preserving expert oversight [30]. PRISM suggests that task-specific models over real oncology EHRs can compete with larger general models for semantic trial matching [68]. Multi-agent and tool-using approaches such as MDAgents, AgentClinic, MedAgentBench, and AI co-scientist systems point toward agentic medical and biomedical workflows [34] [67] [26] [16]. These systems externalize memory and action, but they also expand the safety surface: irrelevant retrieval, poisoning, adversarial prompting, invalid

EHR actions, and multimodal prompt injection must be evaluated as system-level risks [90] [2] [89] [17] [26].

### 3.5 5. Clinical Reasoning and Diagnostic Assistance

Diagnostic assistance is the most attention-grabbing and most easily overstated application. GPT-4 performed strongly on complex diagnostic challenges and USMLE-style tasks, but these studies used curated cases or exam materials [53] [32]. AMIE moved beyond static QA by optimizing for diagnostic dialogue and showing strong performance against primary care physicians in simulated text-chat OSCE-style encounters [82]. AMIE's differential-diagnosis work also suggests that LLM assistance can improve clinicians' lists for difficult NEJM case reports [50].

However, clinical reasoning is more than naming a diagnosis. Nature Medicine evidence on clinical decision-making limitations shows that LLMs may underperform on investigation and treatment choices even when they appear knowledgeable [33]. A randomized trial of LLM influence on diagnostic reasoning adds a crucial human-factor layer: clinician benefit cannot be inferred from standalone model accuracy, because use of the tool depends on training, prompting, trust, and workflow [15]. The defensible conclusion is assistive potential, not replacement.

### 3.6 6. Communication, Documentation, and Administrative Workflows

Communication and documentation are the most immediate deployment zones because they naturally support clinician review. A JAMA Internal Medicine study found that evaluators preferred ChatGPT responses to public patient questions more often than physician forum replies, with higher perceived quality and empathy; nevertheless, the public-forum setting without EHR context limits clinical validity [3]. Patient-message studies move closer to real workflows: LLM-drafted replies and patient-inbox deployments show feasibility but also reveal the absence of stable reference standards for what a good patient reply should be [71] [72].

Documentation tools are similarly promising and risky. Ambient scribes may reduce documentation burden and can generate notes of acceptable quality under blinded review, but quality assurance, billing implications, downstream safety, and clinician review remain concerns [64]. New randomized and pragmatic trials strengthen the deployment evidence: ambient AI scribes are now being evaluated for clinical-practice implementation, work exhaustion, professional fulfillment, utilization, and documentation burden rather than only note text quality [47] [79]. Discharge summarization is attractive because it compresses long hospital stays into handoff documents, yet missing information can be safety-critical [37]. Administrative coding illustrates the same pattern: general LLMs can struggle with ICD extraction, while domain-specific tuning may improve robustness under clinical text variation [5] [45]. These tasks are not trivial paperwork; they shape care continuity, reimbursement, quality measurement, and patient understanding.

### 3.7 7. Multimodal and Generalist Medical AI

Healthcare is multimodal: images, waveforms, lab values, reports, genomics, and longitudinal notes interact. LLaVA-Med, Med-Flamingo, Med-Gemini, and BiomedGPT represent attempts to extend LLM-style interfaces to biomedical images and mixed-modality reasoning [42] [51] [81] [98]. The data layer is equally important: MIMIC-CXR, CheXpert, VQA-RAD, PathVQA, SLAKE, PMC-VQA, and OmniMedVQA anchor much of the medical VLM and VQA evaluation ecosystem [31] [24] [38] [21] [43] [97] [22]. Newer systems such as RadFM, CheXagent, CXR-LLaVA, HuatuoGPT-Vision, and GMAI-MMBench extend this ecosystem toward radiology foundation models, chest-X-ray report assistance, open CXR LVLMs, medical visual instruction tuning, and broad multimodal evaluation [87] [9] [74] [10] [76].

The generalist medical AI perspective argues that foundation models may eventually support flexible, multimodal, task-spanning medical systems rather than isolated classifiers [52]. A scoping review likewise identifies a shift from text-only LLMs toward multimodal AI for documentation, decision support, report generation, and discovery [6]. Multimodality raises the evidentiary bar. Image understanding benchmarks may use simplified VQA pairs rather than full clinical imaging workflows, while radiology report-assistance studies require reader workflow and quality assessment [38] [9]. Long-context and structured EHR tasks require temporal structure and local validation [85] [60]. Multimodal prompt injection is not hypothetical: oncology vision-language systems can be vulnerable to malicious embedded instructions [17]. Thus, multimodal medical LLMs should be evaluated by modality, task, setting, and user role, not treated as a single capability upgrade.

### 3.8 8. Safety, Equity, Security, Ethics, and Governance

Safety evidence cuts across all groups. Targeted misinformation attacks show that medical facts can be corrupted in model behavior while routine benchmark performance remains intact [88]. Data-poisoning studies show that low-level malicious medical misinformation in training corpora can affect targeted medical topics [2]. Adversarial prompting and prompt injection extend these risks to medical QA, patient-summary, and vision-language settings [89] [17]. Hallucination-specific benchmarks such as MedHallBench and MedHallu add domain-specific tests for medical factuality, abstention, and error scoring [99] [59]. Privacy work adds another safety layer: official HIPAA de-identification guidance distinguishes legal de-identification routes from technical anonymity, while PHI benchmarks, annotation frameworks, and clinical-note privacy studies show that LLM-enabled workflows need explicit leakage, over-redaction, membership-inference, and re-identification evaluation at handoff boundaries [54] [73] [78] [65] [27].

Equity evidence is equally central. Health-equity toolboxes and bias studies show that average accuracy can mask subgroup harms, stereotype reproduction, or inequitable recommendations [61] [94]. Ethical reviews distinguish clinician-facing from layperson-facing uses and repeatedly emphasize privacy, misinformation, bias, accountability, and human oversight [19]. Governance documents from WHO, FDA, and NIST provide complementary lenses: WHO frames large multimodal models as health-system technologies needing stakeholder participation and post-release audit; FDA materials highlight lifecycle and premarket considerations for AI/ML medical software; NIST provides cross-sector risk-management structure [57] [13] [55]. These sources are not performance evidence, but they define the obligations that performance evidence alone cannot satisfy.

## 4 Comparative Evidence

| Group | Representative Evidence | Main Contribution | Main Limitation |
|---|---|---|---|
| Biomedical/clinical foundations | BioBERT, PubMedBERT, ClinicalBERT, BioGPT, GatorTron, GatorTronGPT, EHRSHOT, BEHRT, Med-BERT, CEHR-BERT, NYUTron [39] [23] [18] [48] [92] [93] [85] [40] [62] [60] [25] | Domain-specific corpora and structured patient timelines improve biomedical, clinical-note, and EHR prediction tasks. | Mostly benchmark or retrospective prediction evidence; not clinical dialogue or deployment validation. |

| Group | Representative Evidence | Main Contribution | Main Limitation |
|---|---|---|---|
| Medical QA and evaluation | MedQA, PubMedQA, MedMCQA, USMLE studies, Med-PaLM, GPT-4, HealthBench, MedHELM, MedXpertQA, CMExam, MedBench, LiveMedBench [29] [28] [58] [35] [14] [69] [53] [1] [49] [100] [44] [7] [91] | Standardized and live benchmarks measure capability gains, language coverage, contamination risk, and new evaluation dimensions. | Benchmarks remain proxies for clinical care, especially when static, multiple-choice, VQA-style, or model-graded. |
| Open medical models | MEDITRON, MedAlpaca, PMC-LLaMA, ChatDoctor, MedGemma, BioMistral, Med42, Clinical Camel, HuaTuo/HuatuoGPT, DISC-MedLLM, Apollo [8] [20] [86] [41] [77] [36] [11] [12] [80] [83] [96] [75] [84] [63] | Open models improve reproducibility, inspectability, and local validation options. | Openness does not solve data quality, contamination, safety, language coverage, or regulatory validation. |
| Grounded/workflow systems | Almanac, RAG benchmark, TrialGPT, PRISM, MDAgents, AgentClinic, MedAgentBench [95] [90] [30] [68] [34] [67] [26] | Retrieval, ranking, tools, and agents can make LLM outputs more traceable or workflow-specific. | Retrieval errors, invalid actions, adversarial prompting, and prompt injection become first-class system risks. |
| Clinical reasoning | GPT-4 challenges, AMIE, clinical decision critique, diagnostic RCT [32] [82] [50] [33] [15] | LLMs can assist diagnosis in controlled simulations and some clinician workflows. | Routine clinical benefit and safety require prospective, workflow-aware studies. |
| Communication/documentation | Patient Q&A, patient inbox drafts, ambient scribes, discharge summaries, coding [3] [72] [64] [47] [79] [37] [5] [45] | Near-term reviewed workflows can reduce burden and improve drafting under human review. | Quality assurance, readability, coding errors, downstream safety, and accountability remain unresolved. |
| Multimodal/generalist AI | LLaVA-Med, Med-Flamingo, Med-Gemini, BiomedGPT, MIMIC-CXR, CheXpert, VQA-RAD, OmniMedVQA, RadFM, CheXagent, HuatuoGPT-Vision, GMAI-MMBench [42] [51] [81] [98] [31] [24] [38] [22] [87] [9] [10] [76] | Moves from text-only assistance toward real clinical data complexity and broad medical visual evaluation. | Modality-specific validation, security, and clinical integration lag behind capability claims. |
| Safety/governance | Misinformation, poisoning, prompt injection, hallucination, PHI/privacy, equity, ethics, WHO/FDA/NIST/HHS [88] [2] [89] [17] [99] [59] [73] [78] [54] [65] [27] [61] [19] [57] [13] [55] | Establishes the need for security, equity, privacy, lifecycle, and regulatory controls. | Mitigation evidence is still less mature than risk identification and benchmark evaluation. |

## 5 Main Findings

First, medical LLM capability has improved faster than clinical evidence quality. Med-PaLM, Med-PaLM 2, GPT-4, HealthBench, and recent benchmark suites show substantial progress in medical QA, health-response quality, and task coverage [69] [70] [53] [1] [49] [100]. But systematic evaluation literature finds that many healthcare LLM studies still rely on accuracy metrics and limited real patient data, with relatively sparse fairness and robustness evaluation [4]. LiveMedBench further highlights static benchmark contamination and temporal mismatch as evaluation threats [91]. The field therefore risks mistaking an expanding benchmark ecosystem for a mature clinical evidence base.

Second, domain specialization remains useful but not decisive. Biomedical, clinical-note, and structured-EHR pretraining improve tasks that depend on biomedical literature, EHR language, or longitudinal patient trajectories [48] [92] [93] [85] [40] [62] [60] [25]. Open medical LLMs show that curated guidelines, textbooks, PubMed/PMC articles, medical instruction data, doctor dialogues, and multilingual corpora can improve open-model medical-task performance [8] [20] [86] [36] [96] [84]. Yet GPT-4 demonstrates that a general-purpose frontier model can perform strongly on some medical benchmarks without medical-specific training [53]. The practical design question is not simply general versus medical; it is which mixture of base capability, domain data, grounding, local validation, and oversight matches the task.

Third, the strongest near-term deployment case is human-reviewed assistance. TrialGPT and PRISM support trial screening, patient-message systems draft replies, ambient scribes draft notes, CheXagent-like radiology systems draft reports, and coding systems assist constrained administrative extraction [30] [68] [72] [64] [9] [45]. Randomized and pragmatic ambient-scribe trials now add workflow and well-being outcomes to the evidence base [47] [79]. These applications are attractive because they insert LLMs into workflows where humans can review, edit, and reject outputs. The risk is that review becomes nominal rather than meaningful, especially under time pressure, automation bias, or organizational pressure to reduce labor.

Fourth, diagnostic and agentic assistance remain promising but not settled. AMIE's simulated diagnostic dialogue and differential-diagnosis studies are technically impressive [82] [50]. GPT-4 and Med-PaLM 2 show strong medical reasoning under benchmark and challenge conditions [70] [53] [32]. However, clinical decision-making studies, randomized clinician trials, AgentClinic, and MedAgentBench caution that the harder question is not whether an LLM can produce a plausible diagnosis, but whether it improves clinician reasoning, asks for the right information, supports appropriate investigation and treatment choices, and performs valid actions in realistic workflows [33] [15] [67] [26].

Fifth, safety must be treated as an engineering and governance problem, not a disclaimer. Hallucination is only one failure mode. Poisoned data, targeted misinformation, adversarial prompting, prompt injection, unfair recommendations, PHI leakage, re-identification risk, and unclear regulatory scope can all produce harm even when average benchmark performance looks good [88] [2] [89] [17] [61] [94] [73] [54] [65] [27] [13]. A safe medical LLM system needs data provenance, retrieval provenance, output monitoring, equity testing, cybersecurity controls, clinical escalation paths, user training, and lifecycle governance [57] [13] [55].

## 6 Limitations of the Evidence Base

The literature is uneven. Medical QA is overrepresented because it is easy to benchmark, while real-world clinical workflows are harder to study. Many studies use proprietary models whose behavior changes over time, making exact replication difficult [53]. Many open models are preprints or technical reports, which are valuable for method coverage but weaker as clinical evidence [8] [20] [77] [36] [12]. Prospective and randomized evidence remains relatively rare, though patient-inbox, diagnostic-reasoning, ambient-scribe, and some workflow studies are beginning to appear [72] [15] [47] [79].

Data representativeness is a persistent limitation. EHR-trained models such as GatorTron, EHRSHOT/CLMBR, CEHR-BERT, and NYUTron are powerful but tied to particular institutions, coding systems, and documentation cultures [92] [85] [60] [25]. Benchmarks such as MedQA, MedMCQA, CMExam, MedBench, and XMedBench-style multilingual evaluations reflect exam settings, language, and national educational contexts [29] [58] [44] [7] [84]. HealthBench includes physician involvement, but any benchmark still encodes choices about cases, rubrics, languages, and what counts as good advice [1]. Equity tools help surface harms, but they do not prove equitable outcomes after deployment [61].

Finally, this survey itself is limited by the rapid pace of the field. It uses a broad search and evidence-card workflow, but it is not a registered systematic review or meta-analysis. It emphasizes technical synthesis across model families, tasks, and risks. New model releases after the search date should be interpreted through the framework developed here: identify the task, data, user, evaluation type, deployment context, and risk controls before accepting capability claims.

## 7 Open Problems

1. Prospective clinical validation. The field needs more prospective, pragmatic, and randomized studies that measure patient outcomes, clinician workload, diagnostic accuracy, communication quality, equity, and unintended consequences.

1. Evaluation beyond static QA. Benchmarks should include sequential decisions, incomplete information, investigation and treatment choices, multimodal data, interactive EHR actions, contamination control, and abstention or hallucination scoring [33] [67] [26] [99] [59] [91].

1. Grounding and provenance. RAG and tools should report source quality, retrieval sufficiency, irrelevant context, robustness, provenance, and failure cases, not only answer accuracy [95] [90].

1. Human-AI interaction. We need evidence on whether LLM access improves clinician reasoning under realistic training and workflow conditions, and how review interfaces affect documentation, well-being, and decision quality [15] [47] [79].

1. Equity and global validity. Evaluation should include demographic subgroup harms, language access, multilingual benchmarks, and local health-system constraints [61] [57] [44] [7] [84].

1. Security, privacy, and lifecycle monitoring. Medical LLMs need defenses against misinformation, poisoning, prompt injection, hallucination, PHI leakage, re-identification, invalid actions, and benchmark contamination, with lifecycle risk management matched to intended use [88] [2] [89] [17] [26] [99] [59] [73] [54] [65] [27] [91] [55].

1. Regulatory clarity. Systems that generate diagnosis, triage, treatment, or patient-specific recommendations may cross from documentation support into medical device-like functions; intended use and risk should drive governance [13].

## 8 Conclusion

LLMs in medicine and healthcare are best understood as a family of technologies at different evidentiary stages. Biomedical, clinical-note, and structured-EHR language models show the value of domain data; Med-PaLM, GPT-4, HealthBench, MedHELM, MedXpertQA, LiveMed-Bench, and global benchmarks show rapid progress and persistent evaluation limits; open medical models show reproducible experimentation but not automatic deployability; AMIE, TrialGPT, patient messaging, ambient scribes, radiology VLMs, and EHR agents show plausible assistive workflows; and safety/governance studies show why capability is not enough. The central conclusion is therefore conditional: LLMs can be useful in medicine when task scope is explicit, evidence is matched to intended use, human oversight is meaningful, privacy and equity are evaluated, de-identification is not mistaken for complete anonymity, and lifecycle governance is built into deployment. Without those conditions, polished medical language can become a risk amplifier rather than a clinical asset.

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